

Budget-Robust Station Selection for Microtransit Under Demand Uncertainty

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1 Introduction

Virtual bus stop (VBS) selection is a key design problem in microtransit systems because selected stations determine service coverage, passenger walking access, and the ability to pool demand efficiently. However, station selection is a strategic decision made before demand is fully realized, while station activation and OD assignment are operational decisions that can adapt after demand is observed. A station set that performs well under nominal demand may therefore be fragile when demand shifts across OD pairs or time periods.

This motivates a robust station selection model. Instead of optimizing only for a fixed estimated demand vector, we allow OD demand to vary within a budgeted uncertainty set. The first-stage decision selects a long-term set of candidate stations, while the second-stage decisions activate stations and assign OD demand under each realized demand pattern. This timing structure protects service coverage against plausible demand deviations while avoiding the excessive conservatism of assuming that all OD pairs simultaneously attain their worst-case values.

We compare the nominal and robust formulations to evaluate whether robustness improves out-of-sample service coverage quality. In particular, we study how the robust model affects assignment quality, high-percentile passenger cost, and the price paid for robustness.

2 Model

2.1 Nominal Model

2.1.1 Decision Variables

$y_j \in \{0, 1\}$	build station j
$z_{js} \in \{0, 1\}$	station j is active in scenario s
$x_{odjks} \in \{0, 1\}$	assign OD pair (o, d) to origin VBS j , destination VBS k in scenario s

Walking distance induces admissibility sets. Let $\Delta(o) \subseteq J$ be the candidate pickup stations within the walking limit of origin o , and let $\Delta(d) \subseteq J$ be the candidate dropoff stations within the walking limit of destination d . Define the admissible station-pair set

$$A_{od} := \{(j, k) : j \in \Delta(o), k \in \Delta(d)\}.$$

Only pairs in A_{od} may be used to serve OD pair (o, d) .

2.2 Nominal objective

The nominal two-stage objective minimizes expected passenger travel cost:

$$\min_{x,y,z} \sum_{s \in S} \sum_{(o,d) \in \Omega} \sum_{(j,k) \in A_{od}} q_{ods} \underbrace{(d_{oj}^{\text{orig}} + d_{dk}^{\text{dest}} + \lambda c_{jk})}_{a_{odjks}} x_{odjks}, \quad (1)$$

where d_{oj}^{orig} is the walking distance from origin o to VBS j , d_{dk}^{dest} is the walking distance from VBS k to destination d , c_{jk} is the in-vehicle travel time between VBS j and k , and λ is a weight that converts routing time to the same unit as walking distance.

2.3 Constraints

$$\sum_{(j,k) \in A_{od}} x_{odjks} = 1 \quad \forall (o, d), s \quad (2)$$

$$x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} \quad \forall (o, d), (j, k) \in A_{od}, s \quad (3)$$

$$z_{js} \leq y_j \quad \forall j, s \quad (4)$$

$$\sum_j y_j = l, \quad \sum_j z_{js} = k \quad \forall s \quad (5)$$

2.4 Making the model robust against demand uncertainty

2.4.1 Uncertainty Set

For each OD pair (o, d) and scenario type s , let $\bar{q}_{ods}$ denote an upper bound on demand, and let $\bar{Q}_s$ denote an upper bound on *total* demand in scenario type s . The uncertainty set is

$$\mathcal{U}_s = \left\{ q_s : \underline{q}_{ods} \leq q_{ods} \leq \bar{q}_{ods} \quad \forall (o, d), \quad \sum_{(o,d) \in \Omega} q_{ods} \leq \bar{Q}_s \right\}. \quad (6)$$

This uncertainty set says two things simultaneously:

- each OD pair must remain within its own historically calibrated range;
- the aggregate demand in the scenario cannot exceed $\bar{Q}_s$.

Observed demand is integer-valued, but the robust inner maximization may still be solved as a linear program. The key point is that the uncertainty set (6) is a bounded knapsack-type polytope. If the bounds $\bar{q}_{ods}$ and $\bar{Q}_s$ are integral, the inner LP has an optimal extreme point that is integral as well. Hence dropping integrality in the inner maximization does not change the worst-case value, even though the uncertainty set is written in continuous form.

2.4.2 Robust Counterpart

The robust model replaces the nominal objective with

$$\min_{y,z} \sum_{s \in S} \max_{q_s \in \mathcal{U}_s} \Phi_s(z_s, q_s), \quad (7)$$

subject to the first-stage constraints (4)–(5), where $\Phi_s(z_s, q_s)$ is the recourse value defined below.

2.5 Recourse value function

The key modeling point is that x is a *recourse* decision. For fixed scenario-active stations z_s and realized demand vector q_s , define the recourse value function

$$\Phi_s(z_s, q_s) := \min_x \sum_{(o,d) \in \Omega} \sum_{(j,k) \in A_{od}} q_{ods} a_{odjks} x_{odjks} \quad (8)$$

subject to

$$\sum_{(j,k) \in A_{od}} x_{odjks} = 1 \quad \forall (o, d), \quad (9)$$

$$x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} \quad \forall (o, d), (j, k) \in A_{od}, \quad (10)$$

with $x_{odjks} \geq 0$. Integrality is not needed here: for fixed (o, d, s) the feasible region is a simplex intersected with bound constraints, so an optimal extreme point selects a single active station pair.

For fixed z_s , this problem decomposes by OD pair. Define the induced minimum assignment cost

$$\begin{aligned} m_{ods}(z_s) &:= \min_{x_{od..s}} \sum_{(j,k) \in A_{od}} a_{odjks} x_{odjks} \\ \text{s.t.} \quad &\sum_{(j,k) \in A_{od}} x_{odjks} = 1 \\ &x_{odjks} \leq z_{js} \quad \forall (j, k) \in A_{od}, \\ &x_{odjks} \leq z_{ks} \quad \forall (j, k) \in A_{od}. \end{aligned} \quad (11)$$

Because all demand coefficients are nonnegative and the recourse constraints are separable across OD pairs, the recourse value can be written as

$$\Phi_s(z_s, q_s) = \sum_{(o,d) \in \Omega} q_{ods} m_{ods}(z_s). \quad (12)$$

For fixed z_s , the inner worst-case demand problem in (7) becomes

$$\max_{q_s \in \mathcal{U}_s} \sum_{(o,d)} q_{ods} m_{ods}(z_s) \quad \text{s.t.} \quad \underline{q}_{ods} \leq q_{ods} \leq \bar{q}_{ods}, \quad \sum_{(o,d)} q_{ods} \leq \bar{Q}_s. \quad (13)$$

This inner problem will be moved into the constraints through an epigraph variable and then dualized.

2.6 Derivation of the robust counterpart

$$\min_{y, z, \eta} \sum_{s \in S} \eta_s \quad (14)$$

subject to (4)–(5) and

$$\eta_s \geq \max_{q_s \in \mathcal{U}_s} \Phi_s(z_s, q_s) \quad \forall s. \quad (15)$$

Using (12), each (15) becomes

$$\eta_s \geq \max_r \left\{ \sum_{(o,d)} m_{ods}(z_s) r_{ods} : \underline{q}_{ods} \leq r_{ods} \leq \bar{q}_{ods} \quad \forall (o, d), \quad \sum_{(o,d)} r_{ods} \leq \bar{Q}_s \right\} \quad (16)$$

Dualizing the inner LP in (16) gives

$$\begin{aligned} \eta_s \geq & \min_{\alpha_s \geq 0, \beta_{ods} \geq 0, \gamma_{ods} \geq 0} \quad \overline{Q}_s \alpha_s + \sum_{(o,d)} \overline{q}_{ods} \beta_{ods} - \sum_{(o,d)} \underline{q}_{ods} \gamma_{ods} \\ \text{s.t.} \quad & \alpha_s + \beta_{ods} - \gamma_{ods} \geq m_{ods}(z_s) \quad \forall (o, d) \end{aligned} \quad (17)$$

By strong duality, (17) is equivalent to the existence of $\alpha_s \geq 0$, $\beta_{ods} \geq 0$, and $\gamma_{ods} \geq 0$ such that

$$\eta_s \geq \overline{Q}_s \alpha_s + \sum_{(o,d)} \overline{q}_{ods} \beta_{ods} - \sum_{(o,d)} \underline{q}_{ods} \gamma_{ods} \quad \forall s, \quad (18)$$

$$\alpha_s + \beta_{ods} - \gamma_{ods} \geq m_{ods}(z_s) \quad \forall (o, d), s. \quad (19)$$

To remove the value function $m_{ods}(z_s)$, introduce epigraph variables t_{ods} and assignment variables $x_{odjks} \geq 0$:

$$\alpha_s + \beta_{ods} - \gamma_{ods} \geq t_{ods} \quad \forall (o, d), s, \quad (20)$$

$$t_{ods} \geq \sum_{(j,k) \in A_{od}} a_{odjks} x_{odjks} \quad \forall (o, d), s, \quad (21)$$

$$\sum_{(j,k) \in A_{od}} x_{odjks} = 1 \quad \forall (o, d), s, \quad (22)$$

$$x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} \quad \forall (o, d), (j, k) \in A_{od}, s. \quad (23)$$

Because the model minimizes η_s , (18) is tight at optimality, so η can be eliminated from the objective.

Robust counterpart MILP. The resulting formulation is

$$\min_{x, y, z, \alpha, \beta, \gamma, t} \sum_s \overline{Q}_s \alpha_s + \sum_s \sum_{(o,d)} \overline{q}_{ods} \beta_{ods} - \sum_s \sum_{(o,d)} \underline{q}_{ods} \gamma_{ods} \quad (24)$$

subject to

$$\alpha_s + \beta_{ods} - \gamma_{ods} \geq t_{ods} \quad \forall (o, d), s, \quad (25)$$

$$t_{ods} \geq \sum_{(j,k) \in A_{od}} a_{odjks} x_{odjks} \quad \forall (o, d), s, \quad (26)$$

$$\sum_{(j,k) \in A_{od}} x_{odjks} = 1 \quad \forall (o, d), s, \quad (27)$$

$$x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} \quad \forall (o, d), (j, k) \in A_{od}, s. \quad (28)$$

plus (4)–(5), with $x_{odjks}, \alpha_s, \beta_{ods}, \gamma_{ods}, t_{ods} \geq 0$.

3 Experiments

We evaluate the proposed model on operational data from a microtransit system in Zhuzhou, China. The raw dataset contains candidate station locations, a directed routing network between stations, and realized trip requests with timestamps and origin/destination stations. Because the full network is already expensive for repeated robust optimization runs, we also construct a 40-station sampled

instance together with a sparser 40-station variant. The sampled instance preserves the original OD support restricted to the retained stations, while the sparse instance further reduces effective OD coverage. This sparse instance is used for the main model comparison because it preserves most of the observed order volume while substantially shrinking the OD support that must be calibrated from limited historical data.

Table 1: Dataset variants for the April calibration window and May backtest window.

Metric	Full Zhuzhou	Sampled Zhuzhou	Sparse Zhuzhou
Stations	84	40	40
Directed segments	6,972	1,560	1,560
April orders	2,741	2,560	2,560
April ODs with orders	473	389	130
April zero-both stations	15	1	6
May orders	31,546	30,100	30,100
May ODs with orders	1,412	949	131
May zero-both stations	11	1	6

3.1 Experimental Design

Our main comparison uses the sparse 40-station instance (`zhuzhou_data_40_sparse_v3`). We optimize on the first three days of April 2025 (April 1–3) and then evaluate the resulting station plans on the full month of May 2025 (May 1–31). This split is intentionally data-poor on the calibration side: in practice, operators often do not observe dense historical demand for the full OD matrix, especially when demand is highly intermittent at the OD-pair level.

We compare a nominal model against the robust total-demand-cap model. In the sparse study, we fix $l = 30$ candidate stations, use a walking limit of 300 seconds, set the in-vehicle time weight to $\lambda = 10$, and vary the number of active stations per scenario as $k \in \{10, 15, 20\}$. For the robust model, we vary the shared calibration quantile $q \in \{0.90, 0.95, 0.99\}$ used for both the per-OD upper bounds and the scenario-level demand cap.

Table 2: Main sparse-network experiment design.

Item	Setting
Calibration window	April 1–3, 2025
Backtest window	May 1–31, 2025
Instance	40-station sparse Zhuzhou network
Model comparison	Nominal vs. robust total-demand-cap
Candidate stations	$l = 30$
Active stations	$k \in \{10, 15, 20\}$
Walking limit	300 seconds
In-vehicle weight	$\lambda = 10$
Robust calibration	$q \in \{0.90, 0.95, 0.99\}$
In-sample orders	2,560
Out-of-sample orders	30,100

3.2 Uncertainty-Set Estimation

The robust model uses four time-of-day scenarios (morning, afternoon, evening, night). For each period s and OD pair (o, d) , we aggregate historical requests into daily OD counts. The lower bounds are fixed at zero,

$$\underline{q}_{ods} = 0,$$

which matches the implemented calibration routine and reflects the fact that many OD pairs are absent on a large fraction of days. The upper bounds $\bar{q}_{ods}$ are estimated as empirical quantiles of the zero-padded daily OD counts within each period. The scenario-level cap $\bar{Q}_s$ is estimated using the same quantile applied to total period demand. In the baseline diagnostic plots below, we use the 90th percentile calibration to illustrate the shape of the uncertainty set; in the sparse comparison, we then repeat the robust runs for $q \in \{0.90, 0.95, 0.99\}$.

Table 3 summarizes the 90th-percentile calibration on the two 40-station variants. The sampled and sparse instances share the same number of stations and directed segments, but they induce very different observed OD support. This matters because the robust calibration routine imputes missing OD upper bounds after estimating quantiles from the observed April demand. As a result, both 40-station variants produce very large box-to-cap ratios, but the sparse instance starts from much thinner observed OD coverage in every period. In both variants, the aggregate cap is binding in every period, but the sparse instance reaches that regime with substantially fewer observed OD pairs.

Table 3: Uncertainty-set diagnostics for the 90th-percentile calibration on the two 40-station variants.

Period	Sampled ODs	Sparse ODs	Sampled ratio	Sparse ratio
Morning	94	80	17.42	17.46
Afternoon	169	103	7.12	7.28
Evening	249	122	2.86	2.97
Night	120	87	13.16	13.31

4 Results

Both experiments are evaluated on the same out-of-sample test set: all 30,100 May orders from the full-OD `zhuzhou_data_40` dataset (949 distinct OD pairs). For the sparse experiment this is a *distributional shift*: the models were calibrated on a sparsified version of April demand that concentrates all trips onto 130 of those 949 OD pairs, while the May test set contains the real demand distribution. For the sampled experiment the train and test OD supports both come from the same dataset, so there is no such shift.

May costs are computed by `scripts/evaluate_full_od_may.jl`, which for each order finds the minimum-cost feasible station pair (j, k) from the active stations of the relevant time-of-day period. April costs are the in-sample equivalents computed with the same method on the sparse calibration orders. Uncovered = May orders for which no active station lies within the 300-second walking limit.

The nominal model has the lowest mean cost in April (in-sample) for every k , confirming it fits the sparse training data well. On the full-OD May test, however, the nominal model leaves 227 orders uncovered at $k = 10$ and 14 at $k = 15$; the smoothed nominal and robust models achieve

Table 4: Sparse-network comparison (2026-05-03_nominal_smoothed_robust_sparse), evaluated on full-OD May orders. Bold = best in column within each k -block.

k	q	Model	April mean cost	May mean cost	April std.	May std.	Uncovered
10	–	Nominal	4446.97	4305.57	851.46	1092.41	227
10	0.90	Robust	4505.20	4363.35	845.97	1100.00	0
10	0.95	Robust	4521.04	4392.62	836.33	1094.39	0
10	0.99	Robust	4507.51	4368.66	840.04	1089.52	0
15	–	Nominal	4182.48	4031.55	881.21	1082.65	14
15	0.90	Robust	4234.00	4080.66	853.99	1059.06	0
15	0.95	Robust	4235.88	4089.67	848.30	1058.01	0
15	0.99	Robust	4236.58	4073.98	855.82	1067.54	0
20	–	Nominal	4136.13	3972.53	901.73	1094.37	0
20	0.90	Robust	4149.90	3985.01	886.97	1082.80	0
20	0.95	Robust	4155.78	3998.38	883.00	1077.17	0
20	0.99	Robust	4148.22	3984.41	887.77	1079.39	0

complete coverage. At $k = 20$ all models cover every order and mean costs converge to within 30 units of each other.

Figures 1 and 2 show the geographic station selections for $k = 10$. The three models build the same 30 stations but differ in which 10 they leave out: the nominal model excludes peripheral stations (31, 175, 133, 117) while the smoothed and robust models include them at the cost of dropping stations from the dense central cluster. The activation plots reveal the operational consequence: nominal activates only within the high-demand core, leaving the sparse outer regions entirely unserved, while smoothed and robust distribute activations more broadly across all four periods.

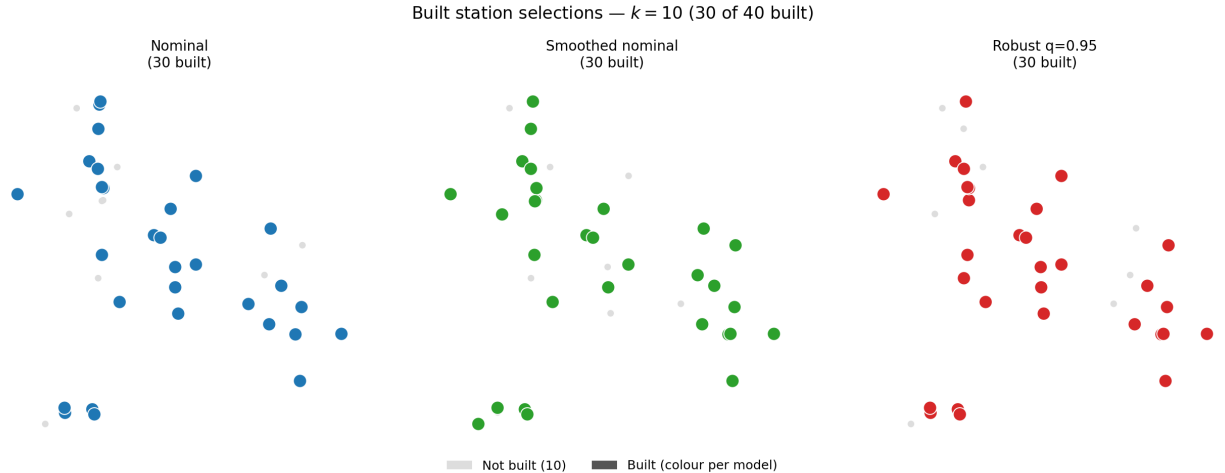


Figure 1: Built station selections for $k = 10$. Coloured dots = built (30 of 40); gray = not built. Nominal excludes peripheral stations that the smoothed and robust models retain.

Figure 3 shows the empirical cost CDFs for $k = 15$ across the four time-of-day scenarios. The distributions for all five models largely overlap, with robust variants shifted slightly to the right in

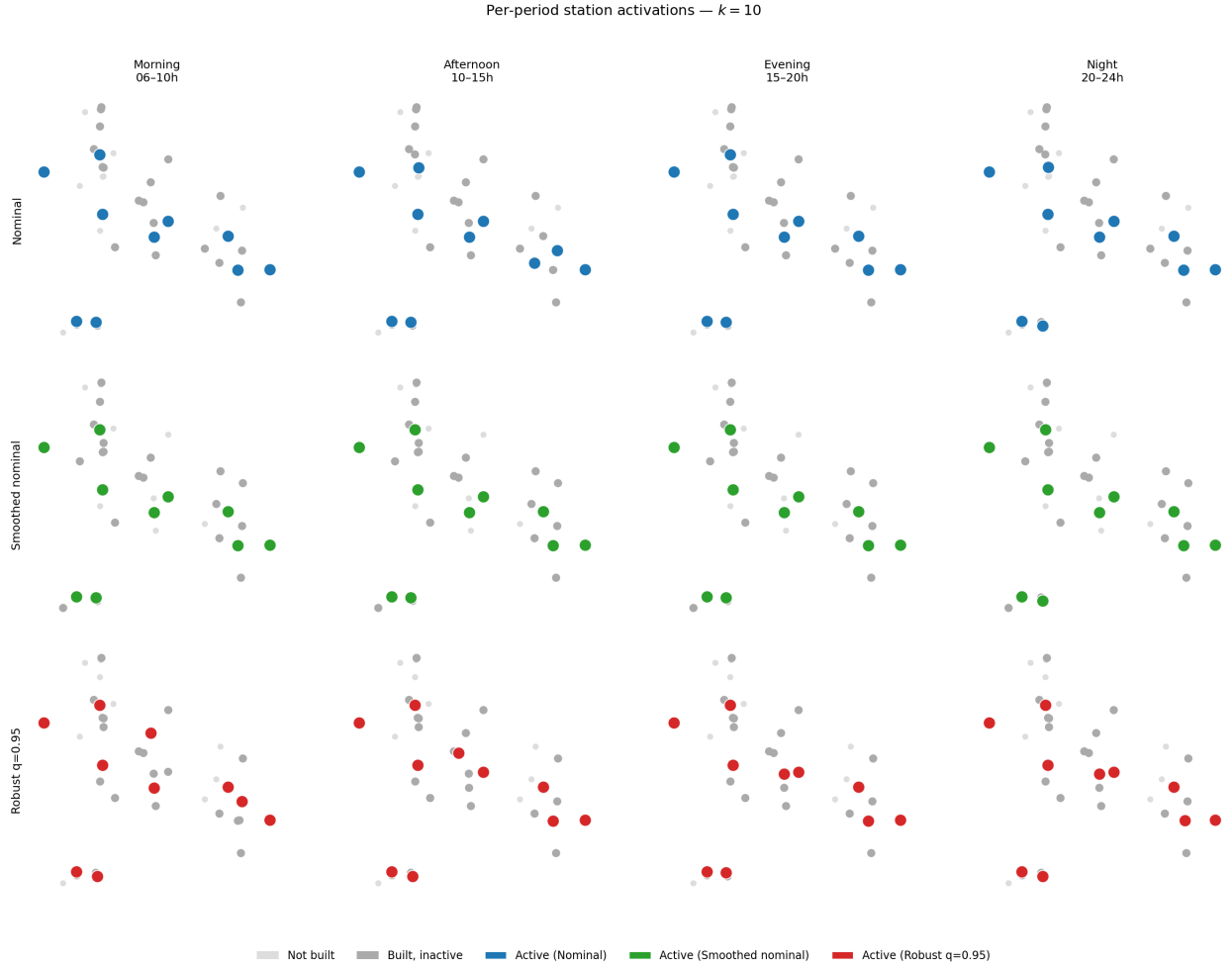


Figure 2: Per-period activations for $k = 10$. Filled = active; hollow = built but inactive; gray dot = not built. Nominal concentrates all 10 active stations in the high-demand core; smoothed and robust spread activations across the network in every period.

the upper tail. The main difference between nominal and non-nominal models is coverage, not the shape of the cost distribution for covered orders.

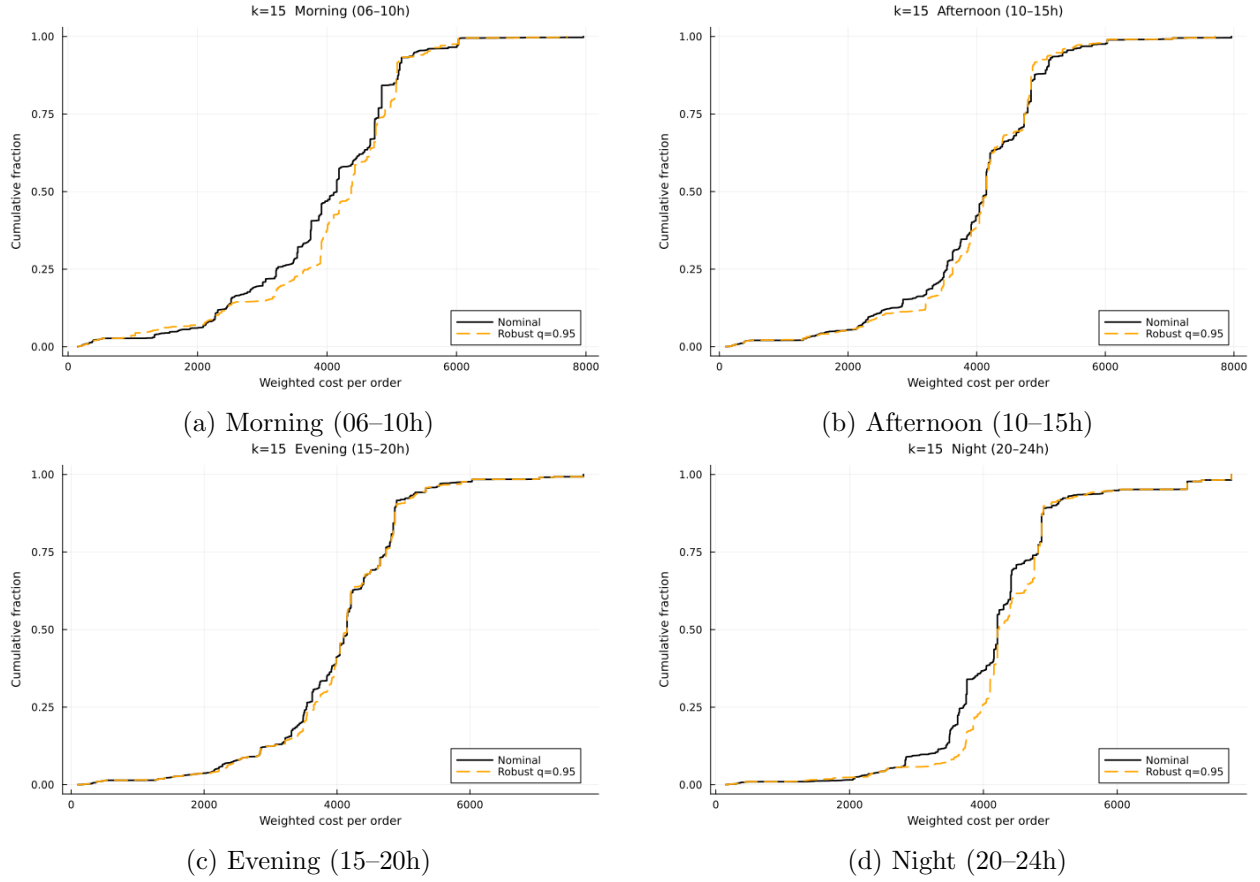


Figure 3: Empirical CDFs of per-order cost for $k = 15$, sparse experiment, May full-OD test set. Each panel corresponds to one time-of-day scenario.

Table 5 reports the same three models on the denser sampled-network instance, where both training and test share the full 949-OD support. All models achieve complete coverage (0 uncovered), and the cost gaps between models are slightly larger than in the sparse setting.

5 Conclusion

Table 5: Sampled-network three-model comparison (2026-05-03_nominal_smoothed_robust), evaluated on full-OD May orders. Bold = best in column within each k -block.

k	q	Model	April mean cost	May mean cost	April std.	May std.	Uncovered
10	–	Nominal	4266.09	4333.42	1129.72	1116.39	0
10	0.90	Robust	4287.81	4359.76	1111.08	1102.67	0
10	0.95	Robust	4302.24	4392.66	1108.89	1090.17	0
10	0.99	Robust	4293.94	4369.71	1109.24	1098.48	0
15	–	Nominal	3977.09	4049.94	1145.30	1119.97	0
15	0.90	Robust	4018.31	4071.27	1106.48	1088.71	0
15	0.95	Robust	4023.38	4080.47	1105.51	1085.45	0
15	0.99	Robust	4030.38	4067.03	1096.25	1066.90	0
20	–	Nominal	3924.95	3984.77	1168.67	1142.49	0
20	0.90	Robust	3967.29	4003.44	1116.51	1082.44	0
20	0.95	Robust	3948.96	3978.06	1118.75	1090.00	0
20	0.99	Robust	3958.28	4005.45	1118.56	1091.22	0